

Lab Assignment no. 2

PRACTICAL NO. 3

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SUBJECT: EDS LAB

3.1.1.NUMPY ARRAY OPERATIONS

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using ndim

- The shape of the array using shape
- The total number of elements in the array using size

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, r and c , representing the number of rows and the number of columns.
- The next r lines contain c space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

Code:

```
import numpy as np

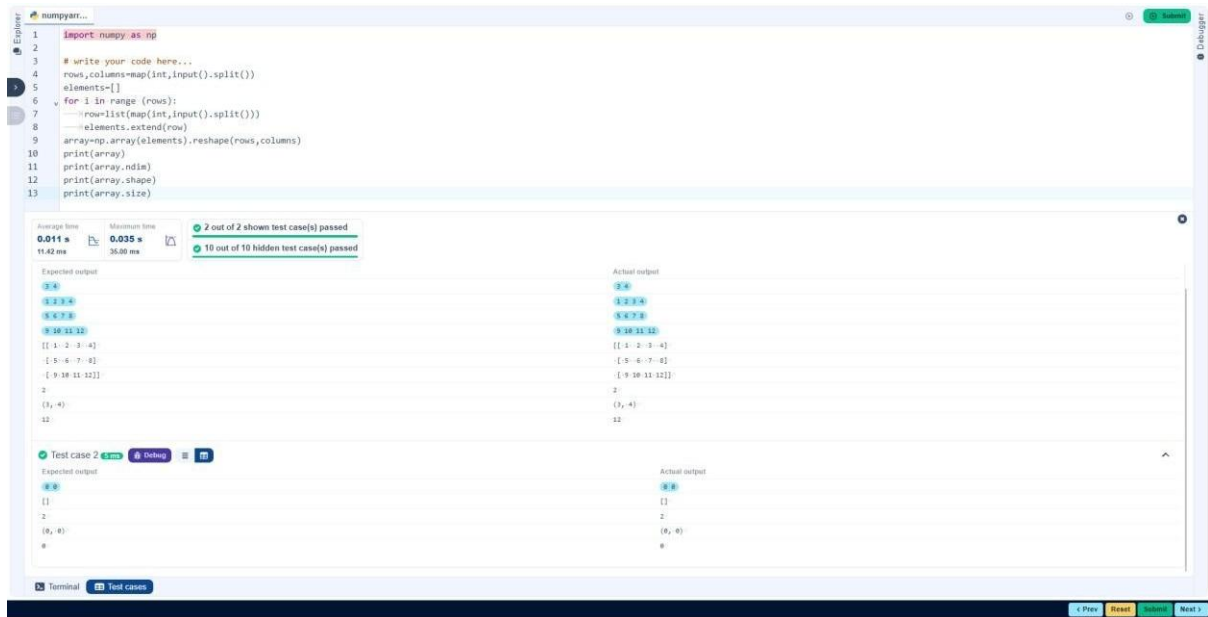
rows,columns=map(int,input().split())

elements=[] for i in range (rows):
    row=list(map(int,input().split()))
    elements.extend(row)

array=np.array(elements).reshape(rows,columns)

print(array) print(array.ndim) print(array.shape)

print(array.size)
```



3.2.1. NUMPY: MATRIX OPERATIONS

The given code takes two 3 x 3 matrices, matrix_a, and matrix_b, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. Addition (matrix_a + matrix_b)
2. Subtraction (matrix_a - matrix_b)
3. Element-wise Multiplication (matrix_a * matrix_b)
4. Matrix Multiplication (matrix_a . matrix_b)
5. Transpose of Matrix A

Input Format:

- The user will input 3 rows for matrix_a, each containing 3 integers separated by spaces.

- Similarly, the user will input 3 rows for matrix_b, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

Code:

```
import numpy as np
```

```
# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])
print("Enter Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])
```

```
# Addition
print("Addition (A + B):")
add=np.array(matrix_a+matrix_b)
print(add) # Subtraction
print("Subtraction (A - B):")
sub=np.array(matrix_a-matrix_b)
print(sub)

# Multiplication (element-wise)
print("Elementwise Multiplication (A * B):")
```

```
multi=np.array(matrix_a*matrix_b) print(multi) #
```

```
Matrix multiplication (dot product) print("A dot B:")
```

```
trans=(matrix_a.dot(matrix_b)) print(trans) #
```

```
Transpose print("Transpose of A:")
```

```
tran=matrix_a.transpose() print(tran)
```

matrixOp...

1 `import numpy as np`

Average time: 0.021 s, Maximum time: 0.029 s

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 (29 ms) Debug

Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
<code>[[1 2 3]</code>	<code>[[1 2 3]</code>
<code>[4 5 6]</code>	<code>[4 5 6]</code>
<code>[7 8 9]</code>	<code>[7 8 9]</code>
Enter Matrix B:	Enter Matrix B:
<code>[[1 1 1]</code>	<code>[[1 1 1]</code>
<code>[1 1 1]</code>	<code>[1 1 1]</code>
<code>[1 1 1]</code>	<code>[1 1 1]</code>
Addition (A + B):	Addition (A + B):
<code>[[2 3 4]</code>	<code>[[2 3 4]</code>
<code>[5 6 7]</code>	<code>[5 6 7]</code>
<code>[8 9 10]</code>	<code>[8 9 10]</code>
Subtraction (A - B):	Subtraction (A - B):
<code>[[0 1 2]</code>	<code>[[0 1 2]</code>
<code>[3 4 5]</code>	<code>[3 4 5]</code>
<code>[6 7 8]</code>	<code>[6 7 8]</code>
Element-wise Multiplication (A * B):	Element-wise Multiplication (A * B):
<code>[[1 2 3]</code>	<code>[[1 2 3]</code>
<code>[4 5 6]</code>	<code>[4 5 6]</code>
<code>[7 8 9]</code>	<code>[7 8 9]</code>
A dot B:	A dot B:
<code>[[6 6 6]</code>	<code>[[6 6 6]</code>
<code>[15 15 15]</code>	<code>[15 15 15]</code>
<code>[24 24 24]</code>	<code>[24 24 24]</code>
Transpose of A:	Transpose of A:
<code>[[1 4 7]</code>	<code>[[1 4 7]</code>
<code>[2 5 8]</code>	<code>[2 5 8]</code>
<code>[3 6 9]</code>	<code>[3 6 9]</code>

matrixOp...

1 `import numpy as np`

Average time: 0.021 s, Maximum time: 0.029 s

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 2 (19 ms) Debug

Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
<code>[[2 4 8]</code>	<code>[[2 4 8]</code>
<code>[9 6 5]</code>	<code>[9 6 5]</code>
<code>[7 8 9]</code>	<code>[7 8 9]</code>
Enter Matrix B:	Enter Matrix B:
<code>[[10 12 13]</code>	<code>[[10 12 13]</code>
<code>[14 15 16]</code>	<code>[14 15 16]</code>
<code>[17 18 19]</code>	<code>[17 18 19]</code>
Addition (A + B):	Addition (A + B):
<code>[[12 16 21]</code>	<code>[[12 16 21]</code>
<code>[23 21 21]</code>	<code>[23 21 21]</code>
<code>[24 26 28]</code>	<code>[24 26 28]</code>
Subtraction (A - B):	Subtraction (A - B):
<code>[[-8 -8 -5]</code>	<code>[[-8 -8 -5]</code>
<code>[-5 -9 -11]</code>	<code>[-5 -9 -11]</code>
<code>[-10 -10 -10]</code>	<code>[-10 -10 -10]</code>
Element-wise Multiplication (A * B):	Element-wise Multiplication (A * B):
<code>[[20 48 104]</code>	<code>[[20 48 104]</code>
<code>[126 90 80]</code>	<code>[126 90 80]</code>
<code>[119 144 171]</code>	<code>[119 144 171]</code>
A dot B:	A dot B:
<code>[[212 228 242]</code>	<code>[[212 228 242]</code>
<code>[259 288 308]</code>	<code>[259 288 308]</code>
<code>[335 366 390]</code>	<code>[335 366 390]</code>
Transpose of A:	Transpose of A:
<code>[[2 9 7]</code>	<code>[[2 9 7]</code>
<code>[4 6 8]</code>	<code>[4 6 8]</code>
<code>[8 5 9]</code>	<code>[8 5 9]</code>

3.2.2.NUMPY: HORIZONTAL VS VERTICAL STACKING OF ARRAYS

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- Horizontal Stacking: Stack the two matrices horizontally (side by side).

- **Vertical Stacking: Stack the two matrices vertically (one below the other).**

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Code:

```
import numpy as np
```

```
# Input matrices
print("Enter Array1:")
arr1 = np.array([list(map(int, input().split())) for i in range(3)])
```

```
print("Enter Array2:")
arr2 = np.array([list(map(int, input().split())) for i in range(3)])
```

```
# Perform horizontal stacking (hstack)
print("Horizontal Stack:")
print(np.hstack((arr1, arr2)))
print("Vertical Stack:")
```

```

1 import numpy as np
2
3 # Input matrices
4 print("Enter Array1:")
5 arr1 = np.array([list(map(int, input().split())) for i in range(3)])
6
7 print("Enter Array2:")
8 arr2 = np.array([list(map(int, input().split())) for i in range(3)])
9
10 # Perform horizontal stacking (hstack)
11 print("Horizontal Stack:")
12 print(np.hstack((arr1, arr2)))

```

Average time: 0.022 s, Maximum time: 0.033 s, 2 out of 2 shown test case(s) passed, 2 out of 2 hidden test case(s) passed

Test case 1

Expected output	Actual output
Enter Array1:	Enter Array1:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Array2:	Enter Array2:
4 5 6	4 5 6
7 8 9	7 8 9
10 11 12	10 11 12
Horizontal Stack:	Horizontal Stack:
[[1 2 3 4 5 6]	[[1 2 3 4 5 6]
[4 5 6 7 8 9]	[4 5 6 7 8 9]
[7 8 9 10 11 12]]	[7 8 9 10 11 12]]
Vertical Stack:	Vertical Stack:
[[1 2 3]	[[1 2 3]
[4 5 6]	[4 5 6]
[7 8 9]	[7 8 9]
[4 5 6]	[4 5 6]
[7 8 9]	[7 8 9]
[10 11 12]]	[10 11 12]]

```

1 import numpy as np
2
3 # Input matrices
4 print("Enter Array1:")
5 arr1 = np.array([list(map(int, input().split())) for i in range(3)])
6
7 print("Enter Array2:")
8 arr2 = np.array([list(map(int, input().split())) for i in range(3)])
9
10 # Perform horizontal stacking (hstack)
11 print("Horizontal Stack:")
12 print(np.hstack((arr1, arr2)))

```

Average time: 0.022 s, Maximum time: 0.033 s, 2 out of 2 shown test case(s) passed, 2 out of 2 hidden test case(s) passed

Test case 2

Expected output	Actual output
Enter Array1:	Enter Array1:
-1 -2 -3	-1 -2 -3
-4 -5 -6	-4 -5 -6
-7 -8 -9	-7 -8 -9
Enter Array2:	Enter Array2:
10 11 12	10 11 12
13 14 15	13 14 15
16 17 18	16 17 18
Horizontal Stack:	Horizontal Stack:
[[-1 -2 -3 10 11 12]	[[-1 -2 -3 10 11 12]
[-4 -5 -6 13 14 15]	[-4 -5 -6 13 14 15]
[-7 -8 -9 16 17 18]]	[-7 -8 -9 16 17 18]]
Vertical Stack:	Vertical Stack:
[[-1 -2 -3]	[[-1 -2 -3]
[-4 -5 -6]	[-4 -5 -6]
[-7 -8 -9]	[-7 -8 -9]
[10 11 12]	[10 11 12]
[13 14 15]	[13 14 15]
[16 17 18]]	[16 17 18]]

3.2.3. NUMPY: CUSTOM SEQUENCE GENERATION

Write a Python program that takes the following inputs from the user:

- **Start value:** The starting point of the sequence.
- **Stop value:** The sequence should end before this value.
- **Step value:** The increment between each number in the sequence.

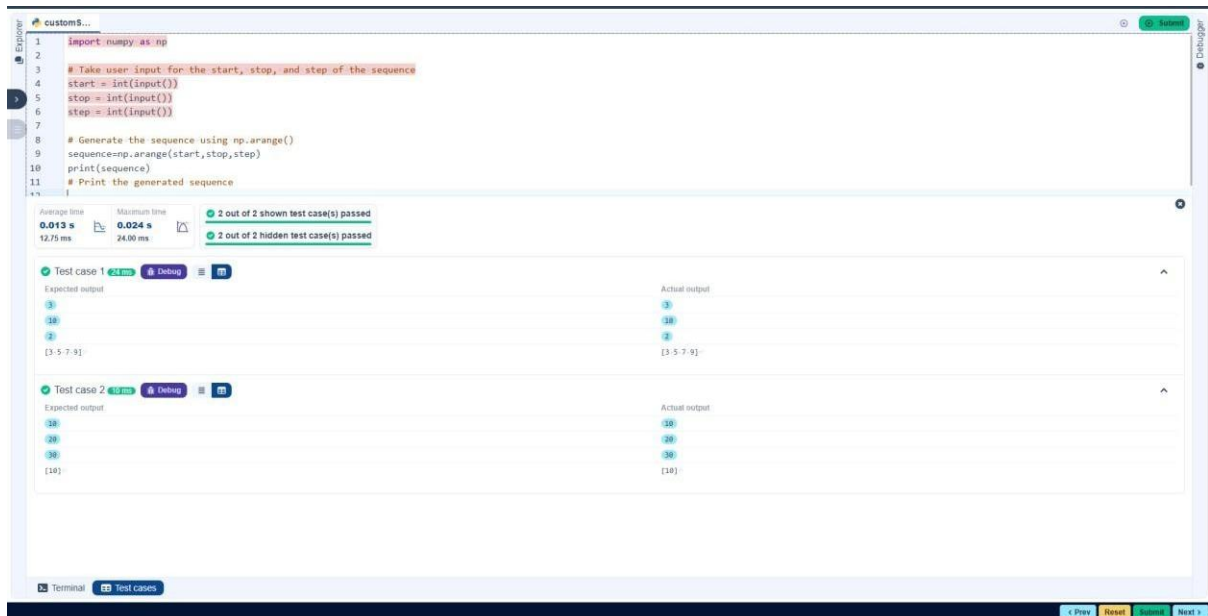
The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

The program should print the generated sequence based on the input values.



3.2.4.NUMPY: ARITHMETIC AND STATISTICAL OPERATIONS, MATHEMATICAL OPERATIONS, BITWISE OPERATORS

You are given two arrays A and B. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: $A_i \text{ OR } B_i$).

Input Format:

- The first line contains space-separated integers representing the elements of array

A.

- **The second line contains space-separated integers representing the elements of array B.**

Output Format:

- **For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.**

Code:

```
import numpy as np
```

```
def array_operations(A, B):
```

```
    # Convert A and B to NumPy
```

```
    arrays x=np.array(A) y=np.array(B) #
```

```
    Arithmetic Operations          sum_result
```

```
    = x+y  diff_result = x-y      prod_result
```

```
    = x*y
```

```
    # Statistical Operations
```

```
    mean_A = np.mean(A)
```

```
    median_A = np.median(A)
```

```
    std_dev_A = np.std(A)
```

```
    # Bitwise Operations
```

```
    and_result = np.bitwise_and(x,y) or_result =
```

```
    np.bitwise_or(x,y) xor_result
```

```
    = np.bitwise_xor(x,y)
```

- # Output results with one space between each element print("Elementwise
Sum:", ' '.join(map(str, sum_result))) print("Element-wise Difference:", '
' '.join(map(str, diff_result))) print("Element-wise Product:", ' '.join(map(str,
prod_result)))

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <original_array>

View array: <view_array>

- After modifying the copy:

Original array after modifying copy: <original_array>

Copy array: <copy_array>

Code: import

numpy as np

```
inputlist = list(map(int,input().split(" ")))
```

```
# Original array original_array =
```

```
np.array(inputlist)
```

```
# Create a view view_array
```

```
=original_array.view()
```

```
# Create a copy copy_array
```

```
=original_array.copy()
```

```
# Modify the view view_array[0] = 99 print("Original array  
after modifying view:", original_array) print("View array:",  
view_array)
```

```
# Modify the copy copy_array[1] = 88 print("Original array  
after modifying copy:", original_array) print("Copy array:",  
copy_array)
```

The screenshot displays the CODETANTRA online IDE interface. The editor shows a Python script that uses NumPy to create an array, a view, and a copy, then modifies them. The script is as follows:

```
1 import numpy as np
2 inputlist = list(map(int, input().split(" ")))
3
4
5 # Original array
6 original_array = np.array(inputlist)
7
8 # Create a view
9 view_array = original_array.view()
10
11 # Create a copy
12 copy_array = original_array.copy()
13
14 # Modify the view
15 view_array[0] = 99
```

Below the editor, the test results are shown. Two test cases are listed, both of which passed. The first test case shows the expected output as a list of integers and the actual output as a NumPy array. The second test case shows the expected output as a list of integers and the actual output as a NumPy array.

Test case 1: Expected output: [10 20 30 40 50 60 70 80]. Actual output: [10 20 30 40 50 60 70 80].

Test case 2: Expected output: [99 2 3 4 5]. Actual output: [99 2 3 4 5].

3.2.6. NUMPY: SEARCHING, SORTING, COUNTING, BROADCASTING

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- **search_value:** The value to search for in the array.
- **count_value:** The value to count its occurrences in the array.
- **broadcast_value:** The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

- 1. Searching:** Find the indices where `search_value` appears in `array1` and print these indices.
- 2. Counting:** Count how many times `count_value` appears in `array1` and print the count.
- 3. Broadcasting:** Add `broadcast_value` to each element of `array1` using broadcasting, and print the resulting array.
- 4. Sorting:** Sort `array1` in ascending order and print the sorted array.

Input Format:

- 1. A single line containing space-separated integers representing `array1`.**
- 2. An integer `search_value` represents the value to search for in the array.**
- 3. An integer `count_value` represents the value to count in the array.**
- 4. An integer `broadcast_value` represents the value to add to each element of the array.**

Output Format:

- 1. The indices where `search_value` occurs in `array1`.**
- 2. The count of occurrences of `count_value` in `array1`.**
- 3. The array after adding the `broadcast_value` to each element.**
- 4. The sorted array.**

Code:

```
import numpy as np
```

```
# Input array from the user array1 =  
np.array(list(map(int, input().split())))
```

```
# Searching search_value = int(input("Value  
to search: ")) count_value = int(input("Value
```

```
to count: ")) broadcast_value =
int(input("Value to add: "))
```

```
# Find indices where value matches in array1
```

```
a=np.where(array1==search_value)[0]
```

```
print(a)
```

```
# Count occurrences in arraycount_nonzero(arrau1==count_value)
```

```
b=np.count_nonzero(array1==count_value)
```

```
print(b)
```

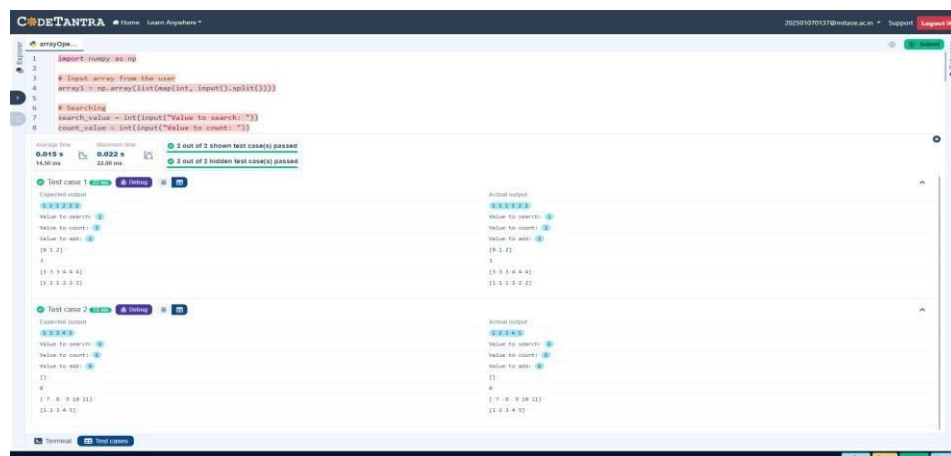
```
# Broadcasting addition c=array1
```

```
+ broadcast_value
```

```
print(c) # Sort the first
```

```
array d=np.sort(array1)
```

```
print(d)
```



3.2.7.STUDENT DATA ANALYSIS AND OPERATIONS

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details:** Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students:** Determine the total number of students in the dataset.

- **Print all student roll numbers:** Extract and print the roll numbers of all students.
- **Print Subject 1 marks:** Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2:** Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3:** Identify the highest marks in Subject 3.
- **Print all subject marks:** Display the marks of all students for each subject.
- **Find total marks of students:** Compute the total marks for each student across all subjects.
- **Find the average marks of each student:** Compute the average marks for each student.
- **Find average marks of each subject:** Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2:** Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3:** Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3:** Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2:** Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2:** Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1:** Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2:** Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored ≥ 90 in each subject:** Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored ≥ 90 :** Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order:** Sort and print the marks of students in Subject 1 in ascending order.
- **Print students who scored between 50 and 90 in Subject 1:** Display students who scored marks between 50 and 90 in Subject 1.

- **Find index positions of students who scored 79 in Subject 1: Identify the index positions of students who scored exactly 79 marks in Subject 1.**

Code:

```
import numpy as np
```

```
a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
```

```
# 1. Print all student details print("All  
student Details:\n", a )
```

```
# 2. print total students
```

```
print("Total Students:",a.shape[0] )
```

```
# 3. Print all student Roll numbers print("All  
Student Roll Nos", a[:,0] )
```

```
# 4. Print subject 1 marks print("Subject  
1 Marks", a[:,1])
```

```
# 5. print minimum marks of Subject 2 print("Min  
marks in Subject 2", np.min(a[:,2] ))
```

```
# 6. print maximum marks of Subject 3 print("Max  
marks in Subject 3", np.max(a[:,3]) )
```

```
# 7. Print All subject marks print("All  
subject marks:", a[:,1:])
```

```
# 8. print Total marks of students print("Total  
Marks", np.sum(a[:,1:],axis=1) )
```

```
# 9. print average marks of each student  
avg=np.mean(a[:,1:],axis=1) print(np.round(avg,1))
```

```
# 10. print average marks of each subject print("Average  
Marks of each subject", np.mean(a[:,1:],axis=0) )
```

```
# 11. print average marks of S1 and S2 print("Average Marks of  
S1 and S2", np.mean(a[:,1:3],axis=0) )
```

```
# 12. print average marks of S1 and S3 print("Average Marks of  
S1 and S3", np.mean(a[:,[1,3]],axis=0) )
```

```
# 13. print Roll number who got maximum marks in Subject 3 i=np.argmax(a[:,3]) print("Roll  
no who got maximum marks in Subject 3", a[i,0] )
```

```
# 14. print Roll number who got minimum marks in Subject 2  
j=np.argmin(a[:,2]) print("Roll no who got minimum marks in  
Subject 2", a[j,0] )
```

```
# 15. print Roll number who got 24 marks in Subject 2
```

```
print(f"Roll no who got 24 marks in Subject 2 [{a[a[:,2]==24,0}]]" )
```

```
# 16. print count of students who got marks in Subject 1 < 40 print("Count of  
students who got marks in Subject 1 < 40", np.sum(a[:,1]<40) )
```

```
# 17. print count of students who got marks in Subject 2 > 90 print("Count of
students who got marks in Subject 2 > 90:", np.sum(a[:,2]>90))

Count=[len(np.argwhere(a[:,1]>=90)),len(np.argwhere(a[:,2]>=90)),len(np.argwhere(a[:,3]>=
90))] count=np.array(Count)

# 18. print count of students in each subject who got marks >= 90 print("Count
of students in each subject who got marks >= 90:", count)

# 19. print count of subjects in which each student got marks >= 90 print("Roll no:", a[:,0]
) print("Count of subjects in which student got marks >= 90:", np.sum(a[:,1:]>=90,axis=1)
)
```

```
(a[:,1]<=90])) k=np.where(a[:,1]==79)
```

```
print(k)
```

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Operation... 0.007 s 7.00 ms 1 out of 1 shown test case(s) passed

Test case 1 7 ms Debug

Expected output	Actual output
All student Details:	All student Details:
[[301, 67, 77, 88]]	[[301, 67, 77, 88]]
[[302, 78, 88, 77]]	[[302, 78, 88, 77]]
[[303, 45, 56, 89]]	[[303, 45, 56, 89]]
[[304, 88, 98, 45]]	[[304, 88, 98, 45]]
[[305, 78, 88, 99]]	[[305, 78, 88, 99]]
[[306, 68, 78, 36]]	[[306, 68, 78, 36]]
[[307, 79, 12, 45]]	[[307, 79, 12, 45]]
[[308, 46, 45, 77]]	[[308, 46, 45, 77]]
[[309, 89, 32, 88]]	[[309, 89, 32, 88]]
[[310, 79, 24, 33]]	[[310, 79, 24, 33]]
Total Students: 10	Total Students: 10
All Student Roll Nos [301, 302, 303, 304, 305, 306, 307, 308, 309, 310]	All Student Roll Nos [301, 302, 303, 304, 305, 306, 307, 308, 309, 310]
Subject 1 Marks [67, 78, 45, 88, 78, 68, 79, 46, 89, 79]	Subject 1 Marks [67, 78, 45, 88, 78, 68, 79, 46, 89, 79]
Min marks in Subject 2 12.0	Min marks in Subject 2 12.0
Max marks in Subject 3 99.0	Max marks in Subject 3 99.0
All subject marks: [[67, 77, 88]]	All subject marks: [[67, 77, 88]]
[[78, 88, 77]]	[[78, 88, 77]]
[[45, 56, 89]]	[[45, 56, 89]]
[[88, 98, 45]]	[[88, 98, 45]]
[[78, 88, 99]]	[[78, 88, 99]]
[[68, 78, 36]]	[[68, 78, 36]]
[[79, 12, 45]]	[[79, 12, 45]]
[[46, 45, 77]]	[[46, 45, 77]]
[[89, 32, 88]]	[[89, 32, 88]]
[[79, 24, 33]]	[[79, 24, 33]]
Total Marks [232, 243, 190, 231, 265, 182, 136, 168, 209, 136]	Total Marks [232, 243, 190, 231, 265, 182, 136, 168, 209, 136]
[77.3 81. 63.3 77. 88.3 68.7 45.3 56. 69.7 45.3]	[77.3 81. 63.3 77. 88.3 68.7 45.3 56. 69.7 45.3]
Average Marks of each subject [71.7 59.8 67.7]	Average Marks of each subject [71.7 59.8 67.7]
Average Marks of S1 and S2 [71.7 59.8]	Average Marks of S1 and S2 [71.7 59.8]
Average Marks of S1 and S3 [71.7 67.7]	Average Marks of S1 and S3 [71.7 67.7]

Terminal Test cases

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Operation... 0.007 s 7.00 ms 1 out of 1 shown test case(s) passed

Average Marks of S1 and S3 [71.7 67.7]

Roll no who got maximum marks in Subject 3 305.0

Roll no who got minimum marks in Subject 2 307.0

Roll no who got 24 marks in Subject 2 [[310]]

Count of students who got marks in Subject 1 < 40 0

Count of students who got marks in Subject 2 > 90: 1

Count of students in each subject who got marks >= 90: [0 1 1]

Roll no: [301, 302, 303, 304, 305, 306, 307, 308, 309, 310]

Roll no: [301, 302, 303, 304, 305, 306, 307, 308, 309, 310]

Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]

Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]

[45, 46, 67, 68, 78, 78, 79, 79, 88, 89]

[45, 46, 67, 68, 78, 78, 79, 79, 88, 89]

[[301, 67, 77, 88]]

[[301, 67, 77, 88]]

[[302, 78, 88, 77]]

[[302, 78, 88, 77]]

[[304, 88, 98, 45]]

[[304, 88, 98, 45]]

[[305, 78, 88, 99]]

[[305, 78, 88, 99]]

[[306, 68, 78, 36]]

[[306, 68, 78, 36]]

[[307, 79, 12, 45]]

[[307, 79, 12, 45]]

[[309, 89, 32, 88]]

[[309, 89, 32, 88]]

[[310, 79, 24, 33]]

[[310, 79, 24, 33]]

[[301, 67, 77, 88]]

[[301, 67, 77, 88]]

[[302, 78, 88, 77]]

[[302, 78, 88, 77]]

[[303, 45, 56, 89]]

[[303, 45, 56, 89]]

[[304, 88, 98, 45]]

[[304, 88, 98, 45]]

[[305, 78, 88, 99]]

[[305, 78, 88, 99]]

[[306, 68, 78, 36]]

[[306, 68, 78, 36]]

[[307, 79, 12, 45]]

[[307, 79, 12, 45]]

[[308, 46, 45, 77]]

[[308, 46, 45, 77]]

[[309, 89, 32, 88]]

[[309, 89, 32, 88]]

[[310, 79, 24, 33]]

[[310, 79, 24, 33]]

(array([6, 9], dtype=int32))

(array([6, 9], dtype=int32))

Terminal Test cases

```
# 20. Print S1 marks in ascending order print(np.sort(a[:,1]))
```

```
print(a[(a[:,1]>=50) & (a[:,1]<=90)]) print(a[(a[:,1]>=45) &
```